

Paper Version Comparison Report

V1 (after_adaptive) vs V2 (ST_yau)
Neural Network Three-Body Problem Research

April 27, 2026

1. Executive Summary

This report provides a comprehensive analysis of all changes between **V1** (Aarush_Paper_v1_after_adaptive.pdf, dated April 10, 2026) and **V2** (Aarush_Paper_v2_ST_yau.pdf, dated April 22, 2026). The two versions are separated by 12 days of revision.

The core finding: **V2 is a precision and conceptual-clarity revision of V1. No new experiments were conducted, no new models were run, and no new results were produced.** The underlying data, figures, tables, and numerical values are identical across both versions. The changes are concentrated in three areas:

- **Title reframing** — the paper’s identity shifts from “neural network emulation” to “hybrid neural force correction”.
- **A factual correction** — ias15 was incorrectly labelled “Symplectic” in V1; V2 corrects this to “Gauss-Radau adaptive”.
- **Terminological tightening in conclusions** — “divergence rate” becomes “error-growth exponent used as a divergence-rate proxy”; language around what SIMON achieves is made more precise.

2. Summary of All Changes — At a Glance

The table below gives a one-line view of every change made in V2 over V1.

Location	V1 (after_adaptive)	V2 (ST_yau)	Type
Title	Neural Network Emulation of the Unrestricted Three-Body Problem	Hybrid Neural Force Correction for the Unrestricted Three-Body Problem	Changed
Date	April 10, 2026	April 22, 2026	Changed
Sec 2 — ias15	Called “Symplectic integrator” that conserves total energy (highlighted as error in PDF)	Corrected to “high-order adaptive integrator based on Gauss-Radau quadrature” — NOT symplectic	Corrected

Location	V1 (after_adaptive)	V2 (ST_yau)	Type
Sec 2 — NN speedup	NN remains constant regardless of t — difference can reach “hundred million times faster”	Hybrid still uses timesteps; speedups depend on architecture and timestep (more nuanced)	Changed
Sec 4.2 — λ term	Uses “Lyapunov exponent as divergence-rate proxy” — yellow highlighted caveat	Cleanly rewritten: “divergence-rate proxy rather than a formal Lyapunov exponent”	Changed
Sec 5 — Concl. 2	Adaptive sub-stepping eliminates non-physical behavior	Adaptive sub-stepping avoids observed non-physical divergence	Changed
Sec 5 — Concl. 3	SIMON bodies “matching ias15”	SIMON bodies “tracking the bounded orbital structure of ias15”	Changed
Sec 5 — Concl. 4	“divergence rate ($\lambda=0.156/\text{yr}$) that indicated preservation”	“error-growth exponent ($\lambda\approx 0.156/\text{yr}$), used as divergence-rate proxy”	Changed
Results (Figs 6–13)	All figures and numerical values	Identical — no new experiments run	No change
References	Same reference list	Identical	No change

3. Detailed Change-by-Change Analysis

3.1 Title Change

V1: *Neural Network Emulation of the Unrestricted Three-Body Problem*

V2: *Hybrid Neural Force Correction for the Unrestricted Three-Body Problem*

This is the most visible change and signals a fundamental reframing:

- **“Emulation” → “Force Correction”:** V1 positioned the work as the NN emulating the simulation outright. V2 narrows the claim to force correction — the NN’s specific role is predicting a correction coefficient to the gravitational force near close encounters, not replacing the integrator wholesale.
- **“Neural Network” → “Hybrid Neural”:** V2 foregrounds the hybrid character of the architecture (NN + ias15 + gating system). SIMON is not a pure NN — it falls back to exact Newtonian gravity 99% of the time.
- **Impact:** The V2 title is more accurate and defensible. A reviewer would have challenged “emulation” since SIMON still uses a step-by-step integrator and only outperforms ias15 at large dt values.

3.2 Date Update

The date was updated from April 10, 2026 to April 22, 2026, consistent with 12 days of revision work. Author names (Aarush Gupta, S. Karthik Yadavalli) are unchanged.

3.3 Section 2 — ias15 Description: Major Factual Correction

This is the most substantively important change. In V1, ias15 was described as follows (the word “Symplectic integrator” was highlighted in yellow in the PDF, indicating the author had already flagged it as incorrect):

V1 (incorrect): *"It is a Symplectic integrator, meaning it is an integrator in which the total energy of the system is conserved. The '15' in ias15 means that it is a 15th order integrator so it can conserve energy well above machine precision. Symplectic integrators have many advantages over non-symplectic integrators, because in the context of chaotic systems (like the Three-Body Problem), they offer higher precision for a relatively larger timestep."*

V2 (corrected): *"It is a high-order adaptive integrator based on Gauss-Radau quadrature, accurate to near machine precision over long integrations [Rein and Spiegel, 2014]. The '15' in ias15 refers to its 15th-order accuracy, which enables excellent energy conservation without being a symplectic method. Its adaptive timestep control makes it especially reliable for close encounters and long-horizon simulations."*

Why this matters:

- **ias15 is NOT symplectic.** Symplectic integrators (e.g., Verlet, leap-frog, WHFAST) preserve the symplectic structure of Hamiltonian systems. ias15 instead uses a 15th-order Gauss-Radau quadrature scheme — it achieves near-machine-precision energy conservation through high accuracy, not through symplectic structure.

- **The V1 description would fail peer review.** Claiming ias15 is symplectic is a verifiable factual error that specialists in numerical integration would catch immediately.
- **The yellow highlight in V1 was a self-correction flag.** This confirms the author knew this needed to be fixed — and V2 has completed that fix.

3.4 Section 2 — Neural Network Speedup Claim: Nuance Added

V1: "...comes with using a Neural Network remains constant, regardless of the temporal distance t from the starting point. This means that the difference in computational efficiency between a Neural Network and an iterative integrator can reach up to a hundred million times faster."

V2: "...comes with using a Neural Network that maps directly from initial conditions to a future state would remain constant, regardless of the temporal distance t from the starting point, which in principle could yield large speedups over step-by-step integrators. In practice, however, the hybrid approach in this work still evolves the system through timesteps, so realized speedups depend on the specific architecture and timestep chosen."

- V1's claim of '100 million times faster' was framed as a general property of NNs emulating the three-body problem — misleading for SIMON since SIMON still steps through time.
- V2 adds the critical caveat that "the hybrid approach in this work still evolves the system through timesteps", correctly limiting the speedup claim to 1.10x–2.09x.
- This is a move from overclaiming to honest scoping — exactly right for peer review.

3.5 Section 4.2 — Divergence Rate: Terminology Cleaned Up

V1 (yellow-highlighted, flagged for revision): "It should be mentioned that the method we are using utilizes the Lyapunov exponent as a divergence-rate proxy, because we are using two different models instead of two perturbed initial conditions under the same dynamics."

V2 (integrated naturally into prose): "Importantly, this quantity is a divergence-rate proxy rather than a formal Lyapunov exponent: we compare two different models rather than two nearby initial conditions evolved under the same dynamics."

- In V1, the caveat was a bolted-on disclaimer (highlighted in yellow), disrupting the flow.
- In V2, the same conceptual caveat is woven cleanly into the analytical prose.
- The term **error-growth exponent** is now used consistently in V2 when referring to $\lambda \approx 0.156/\text{yr}$, making clear it measures model-vs-model divergence, not the system's intrinsic Lyapunov exponent. The numerical value is unchanged.

3.6 Section 5 — Conclusions: Language Precision Upgrades

Three conclusion bullets were reworded for greater precision. The underlying findings are identical.

Conclusion 2 — Adaptive Sub-Stepping

- **V1:** “Adaptive sub-stepping **eliminates** non-physical **behavior** ... allows SIMON to preserve **orbital dynamics**”
- **V2:** “Adaptive sub-stepping **avoids observed** non-physical **divergence** ... allows SIMON to preserve **bounded orbital dynamics**”

Key edits: (a) *eliminates* → *avoids observed* — weaker but more defensible; (b) adds *bounded* — what’s preserved is boundedness, not exact trajectory agreement.

Conclusion 3 — Dominant Failure Mode

- **V1:** “All three bodies ... **matching** ias15”
- **V2:** “All three bodies ... **tracking the bounded orbital structure** of ias15”

Matching implies point-wise agreement. *Tracking the bounded orbital structure* correctly captures that SIMON follows the same bounded region of phase space, but diverges in timing due to chaotic phase drift.

Conclusion 4 — Divergence Rate

- **V1:** “SIMON produced a **divergence rate** ($\lambda = 0.156/\text{yr}$) that **indicated preservation** of the chaotic nature ... without introducing **external instabilities**”
- **V2:** “SIMON produced an **error-growth exponent** ($\lambda \approx 0.156/\text{yr}$), used as a **divergence-rate proxy**, **consistent with** the chaotic nature ... without introducing **catastrophic instabilities over the tested horizon**”

Three changes in one: (a) renames the metric to match Section 4.2 revision; (b) removes an overconfident causal inference; (c) scopes the claim to the test horizon only.

4. Results Comparison — Are the Numbers New?

■ Short answer: No. Every figure, table, and numerical result in V2 is identical to V1. No new models were run, no new experiments were conducted.

Figure/Table	V1 Values	V2 Values	Status
Fig 6 — Trajectory overlay XY	SIMON vs ias15, T=100yr, all 3 bodies	Identical	✓ No change
Fig 7 — Time series overlay	x(t), y(t) for 3 bodies over 100yr	Identical	✓ No change
Figs 8–9 — Additional overlays	Divergence-over-time examples	Identical	✓ No change
Fig 10 — RMS Divergence	$\lambda \approx 0.156/\text{yr}$, window 10–50yr	Identical (λ unchanged)	✓ No change
Fig 11 — Speed-accuracy frontier	dt = 0.005, 0.01, 0.02, 0.04, 0.08	Identical	✓ No change
Fig 12 — dt performance table	Best speedup 2.09x at dt=0.08; lowest RMS 0.847 at dt=0.01	Identical	✓ No change
Fig 13 — Computational cost	23.8 μs –26.3 μs per step, flat profile	Identical	✓ No change
Training metrics	MSE 1.1×10^{-11} , 9s training, NVIDIA RTX 5050	Identical	✓ No change

The PFMN (Pairwise Force Magnitude Network) with 3 hidden layers of 32 neurons using SiLU activations, trained for 5,000 epochs with AdamW (lr 1×10^{-3} , cosine annealing), is the same in both versions. Hardware, hyperparameters, and training/evaluation split are all unchanged.

5. Conclusions Comparison — How the Message Changed

5.1 What Did NOT Change

The following conclusions are textually identical between V1 and V2:

- **Conclusion 1 (SIMON outpaces ias15):** 2.09x speedup at $dt=0.08$, 33ms vs 69ms — identical.
- **Conclusion 5 (What Controls Runtime):** Force evaluation + gating + framework overhead dominates, not integrator math — identical.
- **Conclusion 6 (NN as targeted tool):** NN invoked 1.1% of pair evaluations, 99% exact Newtonian — identical.
- **Conclusion 7 (Optimising implementation):** 54x speed gain from Apple M1 → Nvidia GeForce, architecture changes — identical.
- **Conclusion 8 (Reusable methodology):** Diagnostics give quantitative and qualitative insights for chaotic systems — identical.

5.2 What Changed

Three conclusions were revised, all in the direction of tighter, more defensible scientific language:

#	V1 Conclusion	V2 Conclusion
2	Adaptive sub-stepping ELIMINATES non-physical behavior — preserve orbital dynamics	Adaptive sub-stepping AVOIDS OBSERVED non-physical DIVERGENCE — preserve BOUNDED orbital dynamics
3	Bodies “MATCHING ias15”	Bodies “TRACKING THE BOUNDED ORBITAL STRUCTURE of ias15”
4	“divergence rate” ($\lambda=0.156$) “indicated preservation” — without “external instabilities”	“error-growth exponent” ($\lambda=0.156$) as “divergence-rate proxy”, “consistent with” chaotic nature — without “catastrophic instabilities over the tested horizon”

5.3 Overall Narrative Shift

V1 conclusions read as stronger, more confident claims: the model *eliminates* problems, *matches* ias15, *preserves* the chaotic nature. This language risks overstating what 100-year simulations on a handful of initial conditions can demonstrate.

V2 conclusions pull back to what the evidence actually shows: the model *avoids observed divergence*, *tracks bounded orbital structure*, produces an *error-growth exponent consistent with* chaotic dynamics. These are carefully hedged, scientifically defensible statements that better reflect the diagnostics performed.

This shift — from confident assertion to evidence-bounded conclusion — is the right direction for a paper heading toward peer review or conference submission.

6. Aspects Unchanged — Potential V3 Priorities

For completeness, here are aspects unchanged between V1 and V2 to consider for V3:

- **Section 3.9 (Scope of the Method)** — still frames the work as comparing a “physically constrained NN” to a “traditional iterative integrator”, which is accurate but may benefit from acknowledging that `ias15` is itself non-symplectic.
- **Related Work / Section 1.5** — the Ulibarrena et al. [2024] reference is cited throughout. A dedicated related-work paragraph comparing SIMON more explicitly to their approach would strengthen the positioning.
- **Abstract** — not present in either version; typically required for journal or conference submission.
- **Generalisation of results** — both versions test a single mass configuration (1, 0.01, 0.005 solar masses). A brief discussion of how the approach might generalise to other mass ratios could strengthen the conclusions section.

End of Comparison Report — V1 vs V2 | Neural Network Three-Body Problem Research